

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Introduction

Dynamic pricing adjusts prices based on demand and customer behavior. Thompson Sampling is a Multi-Armed Bandit algorithm that balances exploration and exploitation.

Objective

Design a pricing engine that learns which price maximizes conversions while adapting to changing market conditions.

Problem Statement

Represent different prices as bandit arms, estimate conversion probabilities using Beta distributions, sample from posteriors, choose prices dynamically, and update beliefs after observing customer responses.

Algorithm

1. Initialize $\alpha=1$ and $\beta=1$ for each price. 2. Sample from $\text{Beta}(\alpha, \beta)$. 3. Select highest sample. 4. Observe purchase/no purchase. 5. Update α or β . 6. Repeat and adapt to competitive shocks.

Python Pseudocode

Initialize Beta priors for each price. Repeatedly sample, choose best price, observe reward, update posterior.

Advantages

Learns automatically, adapts quickly, improves revenue, balances exploration and exploitation.

Applications

E-commerce, airlines, hotels, ride sharing, food delivery.

Conclusion

Thompson Sampling provides an efficient adaptive pricing strategy under uncertainty.